Journal of Chromatography A, 1218 (2011) 6987–6994

Contents lists available at ScienceDirect

Journal of Chromatography A

journal homepage: www.elsevier.com/locate/chroma

Effect of pH on protein adsorption capacity of strong cation exchangers with grafted layer

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ARTICLE INFO

Article history:
Received 27 February 2011
Received in revised form 27 July 2011
Accepted 28 July 2011
Available online 6 August 2011

Keywords:
Cation-exchange chromatography
Grafted tentacle layer
Protein adsorption capacity
Ligand density
pH influence
Surface residues
Ionization

ABSTRACT

The effect of pH on the static adsorption capacity of immunoglobulin G, human serum albumin, and equine myoglobin was investigated for a set of five strong cation exchangers with the grafted tentacle layer having a different ligand density. A sharp maximum of adsorption capacity with pH was observed for adsorbents with a high ligand density. The results were elucidated using the protein structure and calculations of pK_a of ionizable groups of surface basic residues. Inverse size-exclusion experiments were carried out to understand the relation between the adsorption capacity and pore accessibility of the investigated proteins.

1. Introduction

Ion-exchange chromatography is the most widely used chromatographic technique for protein separation based on electrostatic interactions between a protein and oppositely charged stationary phase. Since proteins are amphoteric compounds, it is almost always possible to find conditions under which they will bind to a specific ion exchanger. This binding takes place primarily via the surface amino acid residues of proteins having an opposite charge than the adsorbent but other ionic groups, for instance sialyl groups in glycoproteins, can also contribute to the interaction. An advantage of ion-exchange chromatography is the preservation of biological activity of proteins due to mild conditions of elution as well as a high capacity, long lifetime and considerably low cost of many ion-exchange materials [1,2].

Cation-exchange chromatography plays a major role in purification of monoclonal antibodies (MAbs). MAbs belong to the most important pharmaceutical products playing a significant role in diagnostics, research and treatment of many diseases including various types of cancer, autoimmune diseases and allergies [3,4]. Recent development of cell culture technology leads to increasing MAbs titers and puts pressure on the purification processes. Although platform based production allows high production capacities and attractive costs the process often has got to be adjusted to allow purification of larger batches [5]. Cation-exchange chromatography has been widely used as a second capture and polishing step in antibody purification. Nowadays, it is increasingly used also as a capturing step instead of more expensive affinity chromatography [6,7]. Development of new high-capacity cation-exchange resins and proper process optimization are useful for lowering the costs of MAbs production.

The pI value of proteins is often used to predict adsorption conditions for cation-exchange chromatography. The protein net charge is positive at pH lower than its pI value when adsorption on a cation exchanger should occur. The interaction of an adsorbent ligand with a protein is however a more complex phenomenon that cannot be entirely explained by means of total net charge of protein molecule. Examples of protein adsorption on cation exchangers at a pH close to the pI value are also known [8,9]. This phenomenon can be explained by a presence of minor hydrophobic interactions and by uneven distribution of charged patches on protein surface. To properly consider protein surface charge distribution, full three-dimensional structures of proteins should be used.

Hallgren et al. investigated the protein charge influence on the salt-dependent retention factor of Staphylococcal nuclease A [10]. They used mutants with a different number of protonated residues and took into account also uneven charge distribution on the protein surface. Yao and Lenhoff modeled the relation between protein structure and retention on a cation exchanger via interaction free energy [11,12]. These calculations successfully accounted for small

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doi:10.1016/j.chroma.2011.07.097

differences in the retention of cytochrome *c* variants with small structural differences. The approach failed to provide satisfying results in the case of tentacles adsorbents since it did not account for multipoint interactions between the protein and adsorbent and could not predict retention behavior for proteins with significant structural variations either.

Dismer et al. used electrostatic potential calculations for the patches on the lysozyme surface to determine potential binding sites [13,14]. The role of these patches during protein adsorption on cation exchangers was then investigated by means of labeled lysine residues and potential binding orientations on the surface of the adsorbents were proposed. It was shown that protein binding within a tentacle layer led to a high surface coverage and strong retention on the adsorbents. Ishihara et al. proposed an orientation of a monoclonal IgG on a cation exchanger based on the molecular structure and charge distribution of variable regions [15]. The amino acid sequence information was used to successfully predict the retention of various MAbs.

The pore size of a cation exchanger is a parameter of high importance for adsorption design. The pore size must be considered with regard to proteins that are to be separated on the adsorbent. A smaller pore size assures a higher binding surface and potentially also a higher adsorption capacity. It was also declared that smaller pores can provide stronger protein–adsorbent interactions and increased retention due to a more extended contact promoting electrostatic interactions [16]. On the other hand, if the protein size to pore size ratio is too high, the mass transfer resistance is significantly enhanced and the pore accessibility is decreased when large molecules can be even excluded from the pores.

High adsorption capacities of proteins are common for cation exchangers with grafted polymer tentacles despite their relatively small pore sizes [16,17]. Adsorption of proteins takes place here inside a hydrogel layer which can fill nearly the whole pore space. The density of the ligands attached to the tentacles influences the chromatographic performance. An increase in the ligand density enhances the adsorption capacity only in a limited interval of values when a high density may unfavorably decrease pore accessibility. A high ligand density combined with a high protein positive charge resulted in pore clogging due to a high number of protein molecules bound at the pore mouth [18–22].

In this work, we investigated the adsorption performance of a set of five polymethacrylate-based tentacle cation exchangers with varying ligand density. The adsorption capacity was measured for three proteins, polyclonal immunoglobulin G (IgG), human serum albumin (HSA) and equine skeletal muscle myoglobin (Myo), providing model molecules of very different behavior for binding on cation exchangers. Static experiments were carried out at various pH values to elucidate the effect of pH-dependent changes of protein surface charge. Moreover, the pore accessibility of one of the adsorbents was analyzed in regard to the static adsorption capacity of IgG.

2. Materials and methods

2.1. Materials

Five non-commercial, strong ion-exchange materials developed for monoclonal antibody purification by Merck (Darmstadt, Germany) [23] were studied. The adsorbents have a cross-linked polymethyl methacrylate matrix as a support and strong cation-exchange sulfoisobutyl groups as ligands. The ligand density of the adsorbents ranged from 144 μmol/g to 509 μmol/g. The porosity and mean pore radius of these adsorbents were measured by a water desorption method [24]. The porosity was in a range of 0.79–0.81 and the mean pore radius was from 27 nm to 34 nm which means that it was somewhat smaller than the mean pore radius of the adsorbents with the corresponding ligand density used in our previous study [21]. Human normal immunoglobulin G Gammanorm from Octapharma (Stockholm, Sweden), containing IgG_1 (59%), IgG_2 (36%), IgG_3 (4.9%), and IgG_4 (0.5%), was used. The model feed impurities were albumin from human serum (Sigma, MO, USA) and myoglobin from equine skeleton muscle (Sigma, MO, USA).

Chemicals used for buffer preparation (citric acid, dibasic sodium phosphate, and sodium chloride) were of analytical grade. Buffers were prepared using double distilled water and filtered through a 0.45 μm cellulose-nitrate filter (Sartorius, Goettingen, Germany). The pH of the buffers was determined using a calibrated pH electrode (Mettler-Toledo Columbus, OH, USA). Protein solutions were prepared directly before use and were filtered through a 0.22 μm low-protein binding PVDF filter (Millipore, Bedford, MA, USA).

2.2. Static adsorption capacity

The adsorption capacity (often called the static binding capacity in contrast to the dynamic binding capacity) was measured using a 50 mM sodium citrate–phosphate buffer with pH 4–7. A set of experiments was carried out in the buffer containing 75 mM NaCl. The initial protein concentration was either 3 mg/mL or a higher value when the saturation of adsorbent binding sites should be safeguarded for this type of adsorbents [23]. The adsorbents were conditioned in protein-free solutions before each experiment. Extraparticle liquid was removed by suction drying using a glass frit filter with the nominal pore size of 5–10 μm. Wet adsorbent particles in the amount of 0.030 g were then transferred into an Eppendorf vial into which 1.3 ml of a protein solution was added. The Eppendorf vials were placed in a horizontal position on a shaker GFL 1083 (Gesellschaft für Labortechnik, Burgwedel, Germany) and gently stirred at ambient temperature for 15 h. This time was sufficient to reach adsorption equilibrium. The samples were filtered through the low-protein binding filter and analyzed using a flow-through spectrophotometer equipped with a diode array detector at 280 nm or 310 nm (1100 Series LC, Agilent, Palo Alto, CA, USA).

The static adsorption capacity was determined from the decrease of protein concentration in the liquid phase,

$$q = \frac{V_s(c_0 - c^*)\rho_p}{m_p} \tag{1}$$

where q is the static adsorption capacity, V_s the solution volume, c_0 and c^* the initial and equilibrium concentrations of protein in the solution, m_p the mass of wet particles, and ρ_p their density.

2.3. Calculations of pK_a and protonation states of ionizable groups

The H++ server available on http://biophysics.cs.vt.edu/H++ was used to estimate the pK_a values of charged residues in Myo, HSA and IgG_1 at pH 6 [25,26]. The tool adds missing protons to an input protein structure at a given pH and calculates the pK_a values, protonation states and titration curves of ionizable protein groups by means of the Poisson–Boltzmann equation. In order to minimize errors, several structures from Protein Data Bank (http://www.rcsb.org) were run for each of the proteins and the predicted pK_a values were compared. X-ray structures with a high resolution were preferably chosen to increase the accuracy. Pictures of protein surfaces were made by RasMol [27]. The structures used for the final visualization of the surface charged residues were 1e7e for HSA, 2frf for Myo and 1fc1 for the Fc fragment of IgG_1.

Table 1
Characteristics of proteins used.

Protein	p*I*	M_w (kDa)	r_s (nm)
IgG	6.5–10 [32]	150	5.4 [35]
HSA	5.2 [33]	66	3.2 [36]
Myo	7 [34]	17	1.9 [37]

2.4. Inverse size exclusion chromatography

Adsorbent particles were packed into a column using a slurry packing procedure recommended by Merck. Chromatographic experiments were realized in a Pharmacia HR 5/5 column (50 mm × 5 mm i.d., Amersham Biosciences, Uppsala, Sweden). The column was first equilibrated with a mobile phase (50 mM citrate–phosphate buffer with 0, 0.2, or 0.5 M NaCl, pH 5) and probe samples were then injected. The concentration at the column outlet was recorded by means of a refractive index detector (1200 Series LC, Agilent, Palo Alto, CA, USA). The solute retention time in the chromatographic system, t_R, was obtained from the first absolute moment of the residence time distribution function obtained from the chromatographic signal. The distribution coefficient K was then determined from the relationship,

$$K = \frac{V_R - V_{R,min}}{V_{R,max} - V_{R,min}} \quad (2)$$

where V_R is the solute retention volume, $V_{R,min}$ and $V_{R,max}$ are the retention volumes of the largest and smallest solutes used. The experiments were repeated twice.

Glucose (Sigma Chemical, St. Louis, MO, USA), and dextrans (Fluka BioChemika, Buchs, Switzerland and Sigma Chemical, St. Louis, MO, USA) with the weight-average molecular weights of 1200, 6000, 40,000, 56,000, 110,000, 220,000, 490,000 and 3,500,000 were used as solute probes. Their hydrodynamic radius, r_s, was calculated from the Mark–Houwink–Sakurada equation [28,29]:

$$r_s = 0.027 M_w^{0.5} \quad (3)$$

where M_w is the weight-average molecular weight of the solute.

3. Results and discussion

Adsorption of IgG, HSA and Myo on the strong cation exchangers with a grafted polymer layer was investigated in the pH range of 4–7. The cation exchangers are fully ionized at these pH values and the structure of proteins used remains stable. The characteristics of the proteins are given in Table 1. The results of static adsorption experiments in the 50 mM sodium citrate–phosphate buffer are presented in Fig. 1. For the three adsorbents with the lowest ligand density values (144 μmol/g, 236 μmol/g, and 400 μmol/g), the pH dependences of adsorption capacity of all three proteins have the same trends as those observed for both non-grafted and grafted commercial cation exchangers [30,31]. The adsorption capacity had more-or-less the same value in the pH range of 4–5 and strongly decreased above pH 5.

Fig. 1 shows that, for the adsorbents with the ligand density higher than 400 μmol/g, the adsorption capacity of all three proteins exhibited a sharp maximum with respect to pH. The maximum was at pH 5 for IgG and HSA and pH 5.5 for Myo. Fig. 1 also shows that, at pH 4 and 4.5, the adsorption capacity dropped to negligible values in all but one case. This phenomenon of complete pore blocking has also been observed in our previous work focused on the effects of ligand density and ionic strength on the adsorption capacity of IgG [21]. It however occurred at a different threshold ligand density of about 500 μmol/g since a set of particles with larger pores was used.

The phenomenon can be explained by a very high positive charge of proteins at low pH values. The high positive charge combined with a high ligand density may cause a strong charge repulsion effect and exclusion of proteins from pores [21]. At these conditions, a large number of protein molecules bind at the pore mouth preventing more molecules from entering inside the pores [18–22]. Our results confirm that the charge repulsion effect was decisive for the complete exclusion of the proteins whose size differed by the factor of three (Table 1). It is however interesting that the adsorbent with the ligand density of 485 μmol/g provided a non-negligible adsorption capacity of Myo at pH 4.5 (Fig. 1c). This can probably be explained by lower repulsion forces due to the small size of this protein with the hydrodynamic radius of 1.9 nm. The distance of the surface charges of Myo molecules from the adsorbent surface could thus be larger compared to that of IgG and HSA molecules.

As the pH increases, the groups of acidic residues on surface of the molecules become protonated and the overall protein net charge decreases. At pH 5 and 5.5, respectively, the charge repulsion effect faded and very high adsorption capacities were obtained for all three proteins (Fig. 1). The adsorption capacity of IgG increased with the ligand density albeit not proportionally and achieved values as high as 140–150 mg/ml for the two adsorbents with the highest ligand density (Fig. 1a). These values are somewhat lower than those close to 200 mg/ml achieved in our previous work [21].

In the case of HSA, the adsorption capacity was however not proportional to the ligand density at pH 5 and the highest values of about 80 mg/ml were observed for the adsorbents with the medium ligand density (Fig. 1b). On the contrary, the adsorption capacity of Myo was essentially proportional to the ligand density at pH 5 and especially at pH 5.5 (Fig. 1c). These results indicate that the effective accessibility of ligands was here affected by the protein size when a small protein such as Myo blocked fewer binding sites than HSA or IgG.

As has been mentioned above, the adsorption capacity of all proteins decreased sharply at the pH values higher than 5–5.5 (Fig. 1). This decrease was more pronounced for HSA and Myo which adsorption capacity was either negligible or rather small at pH 6 and higher (Fig. 1b and c). On the other hand, the adsorption capacity of IgG at pH 6 reached still remarkable values from about 40 mg/ml to 100 mg/ml and was proportional to the ligand density (Fig. 1a). Thus, the selectivity for IgG binding with respect to HSA and Myo may be expected to increase with pH.

The adsorption capacity of the three proteins in the pH range of 4–7 was measured also in the buffer solution containing 75 mM NaCl (Fig. 2) for the comparison with the results of our previous study [21]. As expected, the presence of the salt had the most pronounced effect at higher pH values where the binding strength is in general weaker. The adsorption capacity of IgG dropped significantly at pH values higher than 5.5 as a result of the competition for the binding sites between the protein and salt ions (Fig. 2a). At pH 6, the adsorption capacities were only 10–20 mg/ml which are about five times smaller values than those obtained in the absence of NaCl (Fig. 1a). At lower pH values, where the protein–ligand interactions are very strong, a salt effect was much less evident. A very mild suppression of charge repulsion effect at pore mouth was observed for IgG at pH 4 and 4.5 for the two adsorbents with the highest ligand densities (Fig. 2a). The effect of protein charge shielding by salt counter-ions was however much weaker than for the adsorbent particles with a larger pore size [21].

The proteins used in this study differ significantly in their charge at a given pH value. The HSA and Myo protein preparations used were homogeneous with the p*I* values of 5.2 and 7, respectively (Table 1). The value of protein p*I* is however an ambiguous parameter for the prediction of protein adsorption on an ion exchanger because this is not determined primarily by the net charge of a pro-

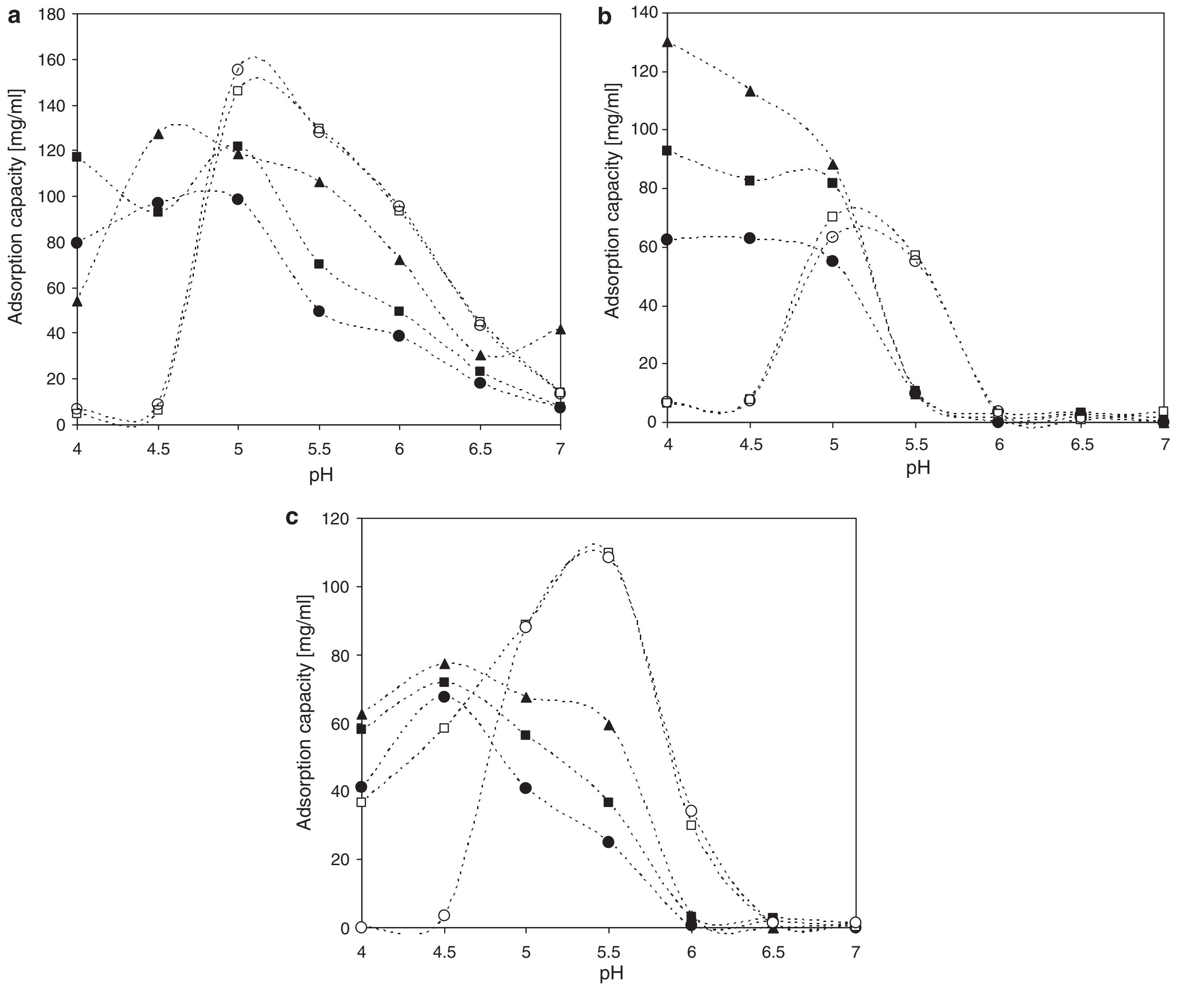

Fig. 1. Adsorption capacity of IgG (a), HSA (b) and Myo (c) vs. pH in the 50 mM sodium citrate–phosphate buffer for five adsorbents with different ligand density values: 144 μmol/g (●), 236 μmol/g (■), 400 μmol/g (▲), 485 μmol/g (□) and 509 μmol/g (○). The connecting lines are made for better visualization of trends.

tein molecule but by the patches of charged groups placed on the protein surface [13,14]. At the pH values close to their respective p*I* values, the adsorption of HSA was significant (Figs. 1b and 2b) whereas the adsorption of Myo was negligible (Figs. 1c and 2c). The HSA adsorption capacity values as high as 90 mg/ml at pH 5 or 50 mg/ml at pH 5.5 imply that that efficient binding of HSA is possible at a protein net charge close to zero. On the contrary, myoglobin did not bind to the tested cation exchangers in significant amounts already at pH 6 or 6.5 when its net charge is still positive.

In order to interpret the pH-dependent adsorption behavior of the proteins, the distribution of surface basic residues and their protonation was analyzed using the H++ tool around the threshold pH, which was assigned to the value above which the protein adsorption capacity was very low or negligible (Table 2 and Fig. 3). The threshold pH values were 6 for HSA and Myo and 7 for IgG. As could be expected, ionizable groups of arginine and lysine side chains were fully protonated in the pH range of adsorption experiments. On the contrary, a portion of histidine residues were nonionized in this pH range since their pK_a-values were lower than the threshold pH value. They could thus be responsible for the loss of interaction ability of entire molecule with the cation exchanger ligand. Table 2 shows that the fraction of nonionized histidine residues at the threshold pH was less than 1.6% of all surface basic residues in the HSA molecule while this fraction was about ten times higher for Myo and Fc fragment of IgG_1. This fact is consistent with that HSA was the only protein which had a significant adsorption capacity at pHs around its p*I* value (Figs. 1 and 2).

Fig. 3 depicts the structures of HSA, Myo and Fc fragment of IgG_1. It is apparent here that some of the nonionized histidine residues together with neighbouring basic residues can create potential binding patches. Direct participation of these patches in protein binding to a cation exchanger would have to be confirmed experimentally, for example, using proteins with introduced point mutations. Nonetheless, the theoretical analysis presented here supported by adsorption capacity results can give some useful ideas about the mechanisms of protein binding.

The polyclonal IgG preparation used in this study contains antibodies with the p*I* between 6.5 and 10 when more than 77% of the mixture form proteins with the p*I* from 8 to 10 [32]. Besides the structure of Fc fragment of the most abundant IgG_1 molecule presented in Fig. 3c, five Fab fragment structures with both κ and λ light chains were analyzed too. Only 0–2 surface histidine residues with pK_a < 7 were found compared to 10 such residues on the Fc fragment (Table 2). It confirms the assumption that the nonionized histidine residues on the Fc fragment played a key role in the pH dependence of IgG adsorption on the investigated cation exchangers.

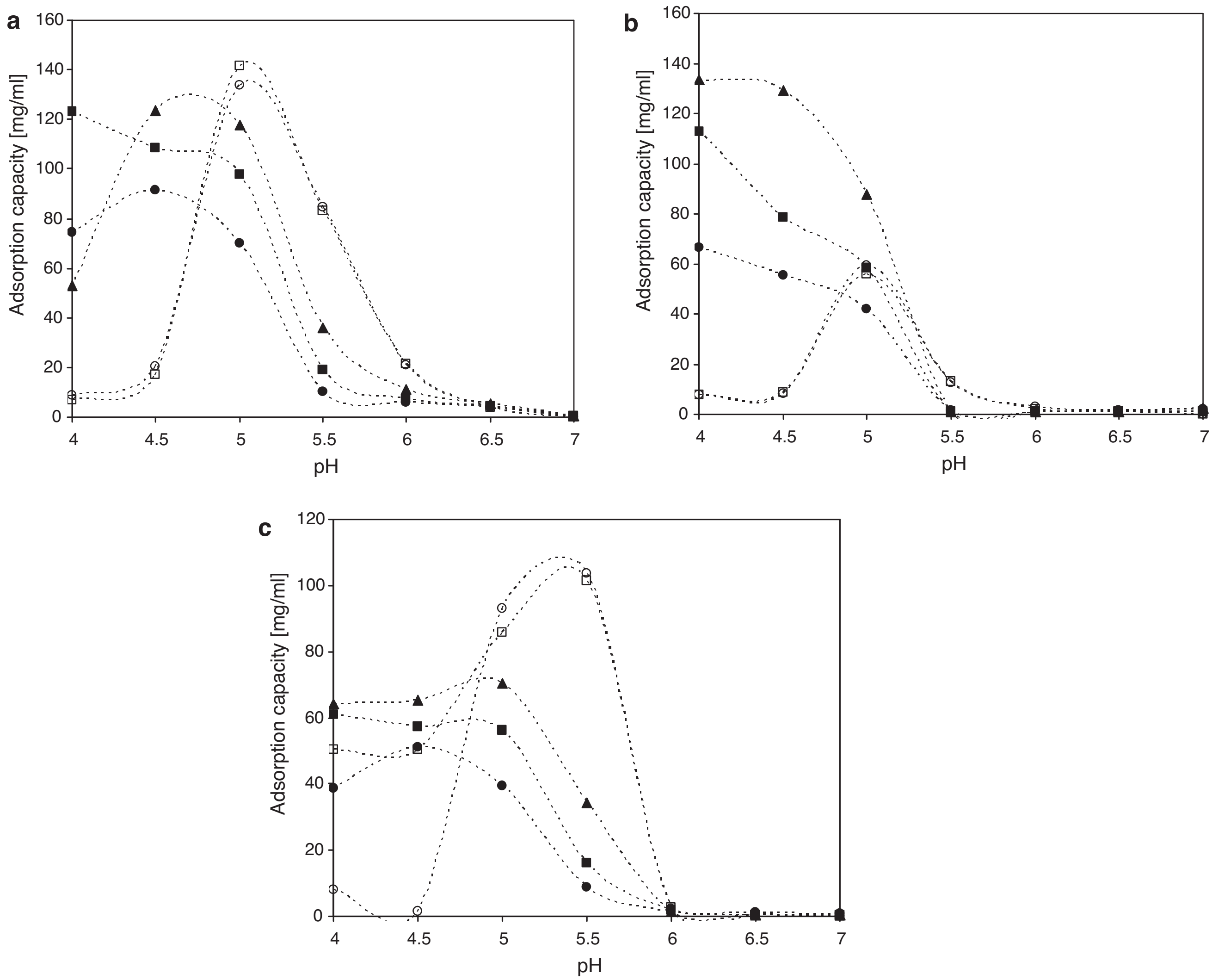

Fig. 2. Adsorption capacity of IgG (a), HSA (b) and Myo (c) vs. pH in the 50 mM sodium citrate–phosphate buffer containing 75 mM NaCl for five adsorbents with different ligand density values: 144 μmol/g (●), 236 μmol/g (■), 400 μmol/g (▲), 485 μmol/g (□) and 509 μmol/g (○). The connecting lines are made for better visualization of trends.

As has been mentioned above, the three proteins used in this study differ significantly in molecular size (Table 1). It is therefore surprising that their adsorption capacities differ relatively little when a significant steric exclusion effect could be expected especially for the adsorption of a large molecule such as IgG on this ion exchanger [23]. We have therefore analyzed the pore accessibility of one of the tested adsorbents, which had a high ligand density of 485 μmol/g and provided IgG adsorption capacity as high as 150 mg/ml. Franke et al. showed that the pore accessibility in this series of adsorbents significantly decreases with the ligand density [23]. They found that only 5% of pore volume of the adsorbent with the ligand density of 485 μmol/g was accessible for IgG molecules at non-adsorbing conditions (50 mM phosphate buffer containing 0.5 M NaCl, pH 7).

The volume of grafted layers of ion exchangers however changes with ionic strength [38]. It was therefore of interest to investigate the accessibility of the adsorbent also at adsorbing conditions when the grafted layer could be swollen. Inverse size exclusion chromatography (iSEC) measurements were performed at three different NaCl concentrations using glucose and a set of dextrans as probe molecules, which encompassed the range of hydrodynamic radius r_s from 0.9 to 49 nm. The adsorbent accessibility for the probe molecules is represented by the partition coefficient K and is plotted in Fig. 4.

Fig. 4 shows that the accessibility of the cation exchanger somewhat increased with the NaCl concentration as could be expected. To quantify this effect in the form of an apparent mean pore size, the data presented in Fig. 4 were fitted with a simplified cylindrical pore model [38]. This model assumes a monodisperse pore size and

Table 2
Analysis of ionization of surface residues of HSA, Myo and Fc fragment of human IgG_1 based on the proteins structure and calculations made by H++ tool.

	HSA	Myo	IgG_1 Fc
Approximate number of surface basic residues	63	26	52
Approximate number of surface histidine residues	10	6	10
Number of histidine surface residues with pK_a below the threshold pH[a]	1	4	10

[a]The threshold pH value was assigned to 6 for HSA and Myo and 7 for IgG.

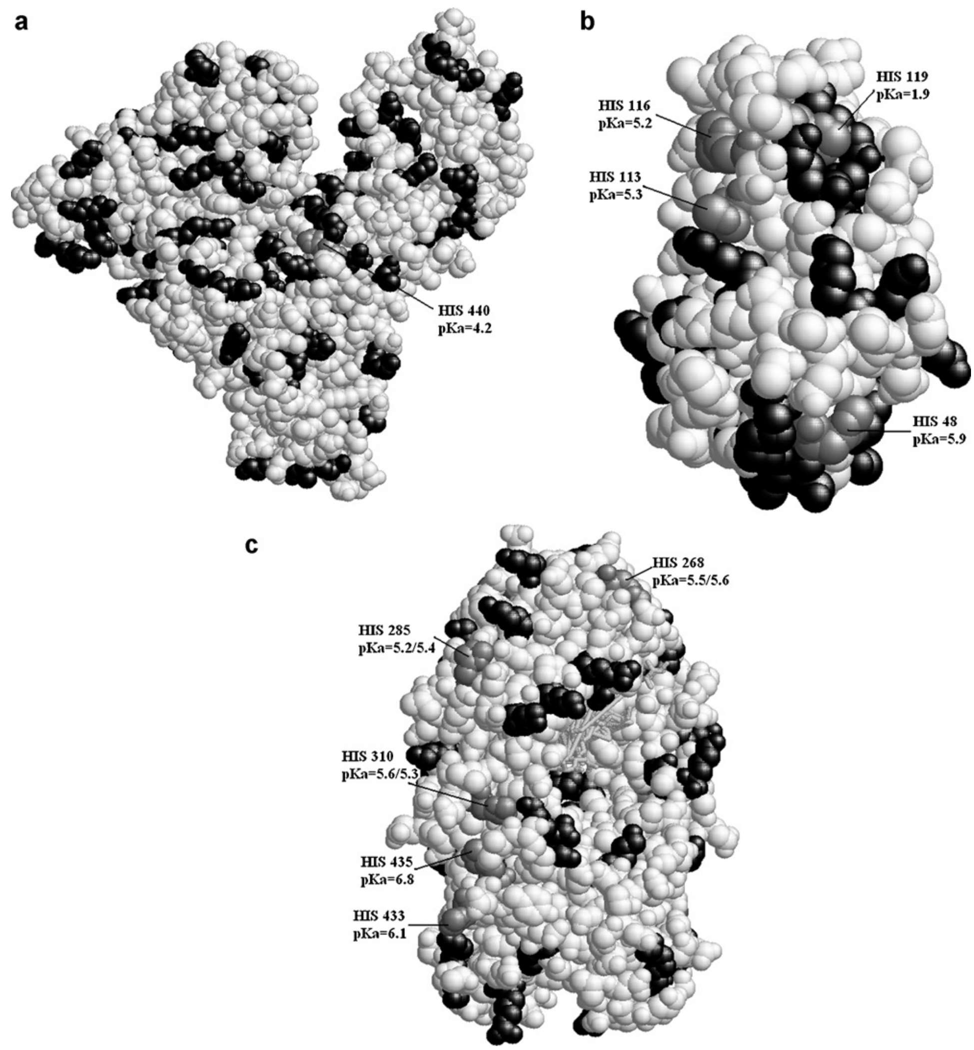

Fig. 3. Three dimensional structure of (a) HSA, (b) Myo, and (c) Fc fragment of IgG_1. The basic residues with pK_a above a threshold pH (6 for HSA and Myo, 7 for IgG) are displayed in black. The residues with pK_a below the threshold pH are displayed in grey and the pK_a values are shown.

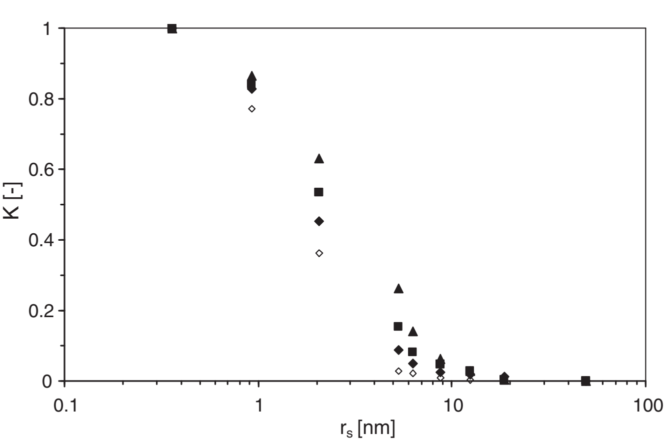

Fig. 4. Partition coefficient vs. dextran hydrodynamic radius for the cation exchanger with the ligand density of 485 μmol/g The symbols represent different NaCl concentrations: 0 M (closed rhombus), 0.2 M (square), and 0.5 M (triangle). Open rhombus represents iSEC experiments carried out for this adsorbent with IgG adsorbed at 0 M NaCl concentration.

approximates the relation between the partition coefficient K and ratio of solute to pore radius, $\Phi = r_s/r_p$, by the following equations:

$$K = (1 - \Phi)^2 \quad \Phi < 1 \tag{4a}$$

$$K = 0 \quad \Phi \geq 1 \tag{4b}$$

The mean pore radius at 0 M, 0.2 M, and 0.5 M NaCl concentration was estimated to be 7.5 nm, 8.7 nm, and 10.7 nm respectively. One interpretation of these results can be that the grafted gel layer shrinks and free pore cross-section increases with the addition of salt [39,40]. The apparent pore size at adsorbing conditions would be then only 70% of the size at non-adsorbing conditions. Another interpretation could be that the effective pore radius is unaffected by buffer ionic strength and the differences in dextran K-values at various salt concentrations origin from unfavourable interactions between uncharged dextrans and negatively bound tentacles [41]. Both these interpretations however must be taken with care because none of them properly accounts for a two-phase character of the pore system composed of the grafted tentacle layer bound to the support surface and liquid phase filling the central part of pore.

The change of the apparent pore size with the salt concentration and small values of the apparent mean pore size indicate quite unequivocally that the grafted layer is essentially not penetrable by the largest dextran molecules. The apparent pore size thus represents a free pore and it does not give any information of the length of the tentacles bound to the adsorbent support. The latter value can be derived from the pore radius value 34 nm obtained by the water desorption method (Section 2.1), Since the tentacle layer collapses during water desorption, this value is close to the radius of the support used for the preparation of these cation exchangers. This would imply that the grafted tentacle layer can be as much as 30 nm thick and can fill over 90% of pore volume.

When the assumed thickness of the tentacle layer containing the ligand and the IgG hydrodynamic radius of 5.4 nm is considered, the protein bound to the ligand molecules of the tentacle layer did not have to reduce the pore cross-section significantly. If the most conservative estimate of IgG specific volume of 0.73 ml/g [42] is taken, the adsorbed amount of about 140 mg/ml of IgG on the cation exchanger with the ligand density of 485 μmol/g (Fig. 1a) would take at least 10% of particle volume, which is about 15% of total pore volume [21]. This volume would form a continuous layer with the thickness of about 3 nm near the support pore wall or of about 16 nm at the other end of the tentacle layer in the contact with the liquid phase filling the pore interior. This layer would be about twice as thick if the cubic packing is transformed into close spherical packing.

This however implies that in average more than one IgG molecule occurs in a radial direction of a tentacle layer cross-section. The mentioned dimensions of the tentacle layer and IgG would allow up to three molecules to be bound next to each other in the radial direction. Since IgG molecules penetrate into the dense tentacle layer in a radial direction from the pore interior toward the support pore wall, they can bind randomly to any ligand placed along a tentacle. IgG molecules bound in the outer third of the tentacle layer would thus interfere in the pore interior. The effective pore size would be then reduced by the adsorbed protein molecules.

To verify this hypothesis, an iSEC experiment for the cation exchanger highly saturated with IgG was performed. Fig. 4 shows that the partition coefficients of dextrans were somewhat lower than those for an empty adsorbent. The calculated apparent pore radius was 6 nm. This 25% reduction of the volume of pore interior corresponds to only about 1% of the total pore volume and 10% of assumed IgG volume. The decrease of pore interior cross-section area could however have a significant impact on the rate of IgG pore diffusion. If Eq. (4a) is applied to the estimation of pore interior accessibility to IgG, the partition coefficient for the empty adsorbent is 0.078 while it is only 0.01 for the adsorbent with 140 mg/ml of IgG bound. It is evident that the low accessibility of pore interior is not an obstacle for achieving a high binding capacity but it would inevitably impede the mass transfer rate from the bulk solution to the adsorbent particles. If the effects of both size exclusion and restricted diffusion are considered, the effective diffusion coefficient would be decreasing by more than one order of magnitude during a chromatographic adsorption process.

4. Conclusions

It was shown that the pH dependence of adsorption capacity of tentacle type adsorbents had the same shape for very different proteins such as immunoglobulin G, human serum albumin and equine myoglobin. At low ligand densities, it was approximately a sigmoidal decreasing function whereas, at high ligand densities, it had a sharp maximum, from which it quickly dropped to negligible values of adsorption capacity within 0.5–2 units of pH. The zero adsorption capacity at higher pH values has a usual cause – prevailing negative surface charges of the proteins and it was observed for all adsorbents independent of their ligand density. On the contrary, the zero adsorption capacity at lower pH values was observed only in the case of high ligand density adsorbents and was a result of effective pore blocking due to extensive binding of proteins at the pore mouth and subsequent electrostatic repulsion of equally charged protein molecules.

It was further found that the pH maximum of the adsorption capacity did not correlate with the p*I* value of the proteins. The difference between these two values was about 3 units for IgG and 1.5 units for myoglobin. On the other hand, the pH maximum of HSA adsorption capacity occurred approximately at its p*I*. These results were successfully interpreted using the calculation of protonation of ionizable groups of basic residues on the protein surface. This emphasized the importance of the distribution of charges on the protein surface and a need for careful pH screening during optimization of cation exchange chromatography of proteins.

Inverse size exclusion chromatography experiments with dextran probes were carried out for an adsorbent with a high ligand density in order to learn why no significant correlation was observed between the adsorption capacity and pore accessibility of these proteins of very different size. It was found that the tentacle layer can occupy over 90% of the total pore volume and its change with the ionic strength has not been unequivocally proven. This volume was sufficiently large to accommodate the adsorption capacity as high as 150 mg/ml. Protein molecules carrying a positive charge could enter the tentacle layer whereas this layer had a very limited accessibility for uncharged larger dextran molecules. The dependence of partition coefficients of dextran probes on their hydrodynamic radius allowed estimating the radius of free-pore interior. At binding conditions, the radius of the free-pore interior was about 40% larger than the hydrodynamic radius of IgG. This difference was reduced to 10% after full saturation of the adsorbent with IgG. The free-pore interior would thus have a very different accessibility for these three proteins which, however, did not affect their adsorption capacity but it could significantly impair their mass transfer.

Acknowledgements

This work was supported by grants from the 6th Framework Program of EU, Project Advanced Interactive Materials by Design (AIMs) no. NMP3-CT-2004-500160 and from the Slovak Grant Agency for Science, VEGA 1/0655/09. We thank our colleagues from Merck KGaA (M. Johnck, M. Schulte) and Universites d'Aix – Marseille I, II et III – CNRS (R. Denoyel, M. Barrande, and I. Beurroies) for providing the materials and data on the ligand density and pore characteristics.

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